TinyTrack

Childcare Enrollment Management Software

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CST-452 Milestone 6

Final Project Completion and presentation

Grand Canyon University

Instructor: Professor Amit Karmaker

Date: 10.7.26

Video Link –
youtube.com/watch?v=uGvw82XP_K8&feature=youtu.be

Git Link –
https://github.com/lindsdeck/TinyTrack

ABSTRACT

Tiny Track is a web-based childcare enrollment management application developed using Java, Spring Boot, Thymeleaf, and MySQL. The application was created to help childcare administrators manage student records, classroom assignments, licensed capacities, current openings, and future enrollment projections from one centralized system.

The completed application allows administrators to manage centers, classrooms, and student enrollment records. It includes student search, add and edit functionality, and inactive record management, live dashboard statistics, automatic projected departure dates, and classroom capacity calculations. TinyTrack also includes an enrollment projection engine that evaluates student age, enrollment dates, projected departure dates, classroom age requirements, and licensed capacities to estimate future classroom placement and availability. A Month to Month classroom planning view was also implemented. This feature displays projected classroom rosters for any selected date, orders students from oldest to youngest, identifies upcoming classroom transitions, identifies students held in younger classrooms because of capacity limitations, and displays enrollments scheduled to begin within the next five weeks.

The system reduces the need for manual spreadsheets and repeated calculations while improving accuracy, organization, and enrollment planning. The completed project demonstrates how software can support better operational and enrollment decisions within childcare administration. The expanded forecasting interface gives administrators a centralized view of classroom movement, upcoming enrollment, capacity restrictions, and students requiring manual review.

History and Signoff Sheet

Change Record

Date	Author	Revision Notes
6.24.26	Lindsey	Initial draft for review/discussion
7.15.26	Lindsey	Final Architecture Plan created. Changes made from instructor feedback
8.2.26	Lindsey	Updates architecture document to reflect completed TinyTrack implementation, testing and final functionality
9.13.26	Lindsey	Added Month to Month classroom forecasting, transition indicators, capacity highlighting, five week previews, and updated documentation
9.27.26	Lindsey	Updated documentation to reflect Milestone 5
10.7.26	Lindsey	Updated final documentation to reflect the implemented application, corrected architecture and class descriptions and added the final requirements review, testing summary, and planned improvements

Overall Instructor Feedback/Comments
Add descriptions for the nonfunctional requirements.

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No Abstract

Overall Instructor Feedback/Comments

Integrated Instructor Feedback into Project Documentation

☒ Yes ☐ No

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Design Planning Summary

Problem and Background

Childcare providers often rely on paper files, spreadsheets, or a combination of both to manage enrollment, classroom capacity, and future openings. While these methods are familiar, they require frequent manual updates and make it difficult to accurately predict classroom availability. As enrollment changes, administrators must constantly calculate future openings based on children's ages, classroom ratios, and anticipated graduation dates. This process is time consuming and increases the likelihood of scheduling errors and missed enrollment opportunities.

As someone who works in childcare administration, I have experienced these challenges firsthand. Existing childcare management software offers many features, but there are very few systems designed specifically to help administrators accurately forecast future enrollment and classroom availability. This often results in providers relying on manual calculations that consume valuable time and can negatively impact enrollment decisions.

TinyTrack was developed to address these challenges by providing a centralized web-based application that automates enrollment management and future classroom forecasting. The application allows administrators to maintain child records, manage classroom capacities, calculate future classroom placement based on age progression, and instantly view projected openings for any selected date. By automating these processes, the system improves efficiency, reduces administrative workload, increases data accuracy, and supports better business decisions.

Overview of Design Concepts

TinyTrack was developed as a web-based application using a layered architecture. The application separated the presentation layer, business logic layer, and data access and storage layer. Controllers handle browser requests, services and the projection engine proves enrollment rules, and repositories retrieve and save information in MySQL. This structure supports maintenance, troubleshooting, and future enhancements.

The presentation layer provides the user interface that administrators interact with through a web browser. It contains the application dashboard, forms, navigation, search features, student management screens, classroom management screens, center information, and future enrollment projections.

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The data layer uses a MySQL database to store childcare centers, classrooms, student records, enrollment dates, projected departure dates, and classroom capacities. Spring Data JPA repositories retrieve and save these records. The current implementation does not store user accounts or provide authentication.

One of the most important concepts of design is the future availability insight. Instead of simply displaying current enrollment, the application projects future classroom availability by evaluating each child's birthdate, enrollment end data, classroom age requirements, and classroom capacity. This feature provides administrators with accurate enrollment projections and supports informed decision making when accepting new families.

The forecasting interface includes a Month to Month classroom view. Administrators select a projection date and review classroom rosters ordered from oldest to youngest. Visual indicators identify approaching classroom transition, students held in younger classrooms because of capacity restrictions, and enrollments beginning within the next five weeks.

System Architecture

The childcare Enrollment Management System follows a four-layer architecture that separates the application into independent components. Each layer performs a specific function while communicating only with the layers directly above or below it. This approach improves readability, maintainability, scalability, and testing.

Presentation Layer

The presentation layer was implemented using Thymeleaf, HTML, and CSS. It provides the dashboard, navigation, student forms, classroom forms, center information, student search, classroom capacity views, and future enrollment projection screens. Future enrollment and Month to Month forecasting tabs, classroom roster tables, transition highlighting, capacity-hold highlighting, upcoming enrollment previews, and a complete future enrollment schedule.

Business Logic Layer

The business logic layer processes requests received form the user interface. It applies all enrollment rules, validates data, determines classroom assignments, performs future availability calculations, and coordinates communication between the presentation layer and the database. Services include:

- StudentService
- ClassroomService

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- CenterService
- DashboardService
- EnrollmentProjectionEngine
- AgeCalculator

Data Layer

The data layer consists of MySQL database that stores childcare centers, classrooms, student information, enrollment dates, projected departure dates, classroom assignments, age ranges, and licensed capacities.

Project Objectives

- Designed and developed a web-based enrollment management system for childcare providers
- Created a centralized MySQL database for managing student enrollment information
- Implemented current and future classroom capacity calculations
- Created a user-friendly interface for childcare administrators
- Developed an enrollment projection engine that supports age-based classroom placement and capacity restrictions

Database Design

Database Overview

The Childcare Enrollment Management System uses a MySQL relational database to store all application data. A relational database was selected because it provides structured storage, maintains relationships between data, enforces referential integrity, and supports efficient searching and reporting. The database is designed to minimize redundant data while allowing administrators to quickly retrieve information related to children, classrooms, enrollment and future availability.

Each major component of the application is represented by its own table. Relationships between tables are established using primary and foreign keys to ensure data consistency. This allows child records, classroom assignments, enrollment history, and

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user accounts to be managed independently while remaining connected throughout the application.

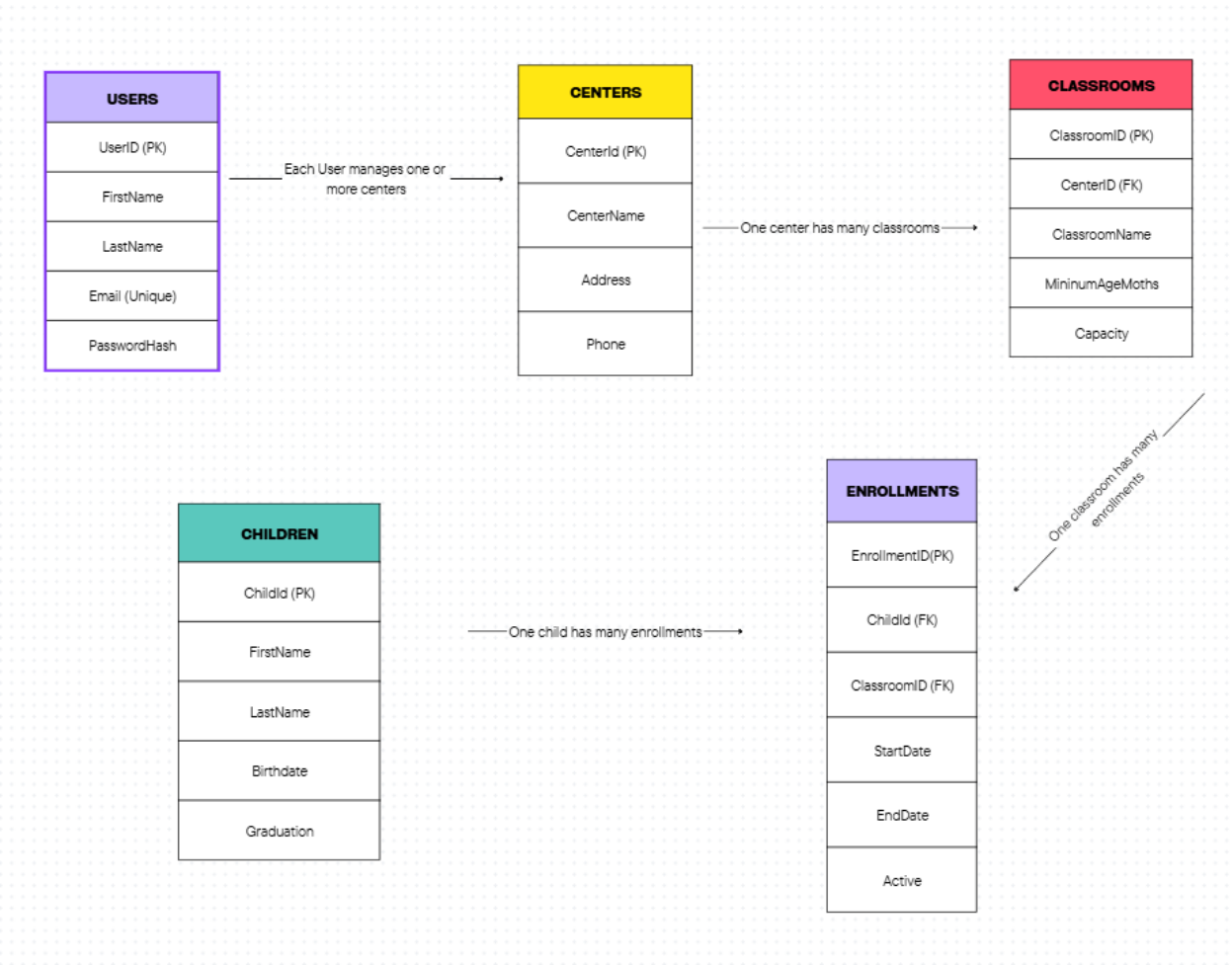
The database also supports the Future Availability feature by storing enrollment dates, anticipated graduation dates, classroom assignments, and classroom capacities. These values are used by the application to calculate projected openings without requiring duplicate data to be stored.

Database Tables

Table	**Purpose**
Centers	Stores childcare center information
Classrooms	Stores classroom information and capacities
Students	Stores child demographic information

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Entity Relationship Diagram



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Table Descriptions

Centers

The centers table stores information about each childcare location. Users may have access to multiple locations.

Field	Type	Key
CenterID	INT	PK
CenterName	VARCHAR(50)	
Address	VARCHAR(100)	
Phone	VARCHAR(20)	
City	VARCHAT(75)	
State	VARCHAR(2)	
Active	Boolean	

Classrooms

Each classroom belongs to one childcare center and contains information about age groups and enrollment capacity

Field	Type	Key
ClassroomID	INT	PK
classroomName	VARCHAR(75)	
MinAgeMonths	INT	
MaxAgeMonths	INT	
Capacity	INT	
Active	Boolean	
CenterId	BIGINT	FK

Students

This stores all the demographic information for all children.

Field	Type	Key
ChildID	INT	PK
FirstName	VARCHAR(50)	
LastName	VARCHAR(50)	
Birthdate	DATE	
EnrollmentDate	DATE	

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ProjectedExitDate	DATE	
Active	BOOLEAN	
ClassroomID	BIGINT	FK

Object and Class Design

The Childcare Enrollment Management System is designed using an object-oriented approach. Each class represents a major component of the application and is responsible for managing a specific type of data or functionality. Separating responsibilities into individual classes improves maintainability, promotes code reuse and makes the application easier to expand in future versions.

The application includes model classes that represent the data stored in the database, service classes that contain business logic, and repository classes that manage communication with the database. Together, these classes provide a structured approach for processing enrollment information, calculating classroom availability, and maintaining data integrity.

Major Classes:

- Center
    - Stores information about each childcare center.
- Classroom
    - Stores classroom information including age group and capacity.
- Student
    - Represents each enrolled child and stores identifying information.
- EnrollmentService
    - Manages child enrollment and classroom assignments.
- EnrollmentProjectionEngine
    - Calculates future classroom availability based on enrollment data.
- ClassroomProjection
    - Stores projected and upcoming students, licensed capacity and calculated openings
- ProjectedStudents
    - Combines a student with the calculated age, projected classroom, age-eligible classroom, placement status and upcoming status.
- ProjectedResults

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    - Organizes classroom projections, future enrollments, excited students, unplaced students, and projection warnings for a selected date.
- ProjectedStatus
    - Identifies if the student has a normal placement, will move soon or was unable to move.

Class Relationships

A User manages one or more childcare centers.

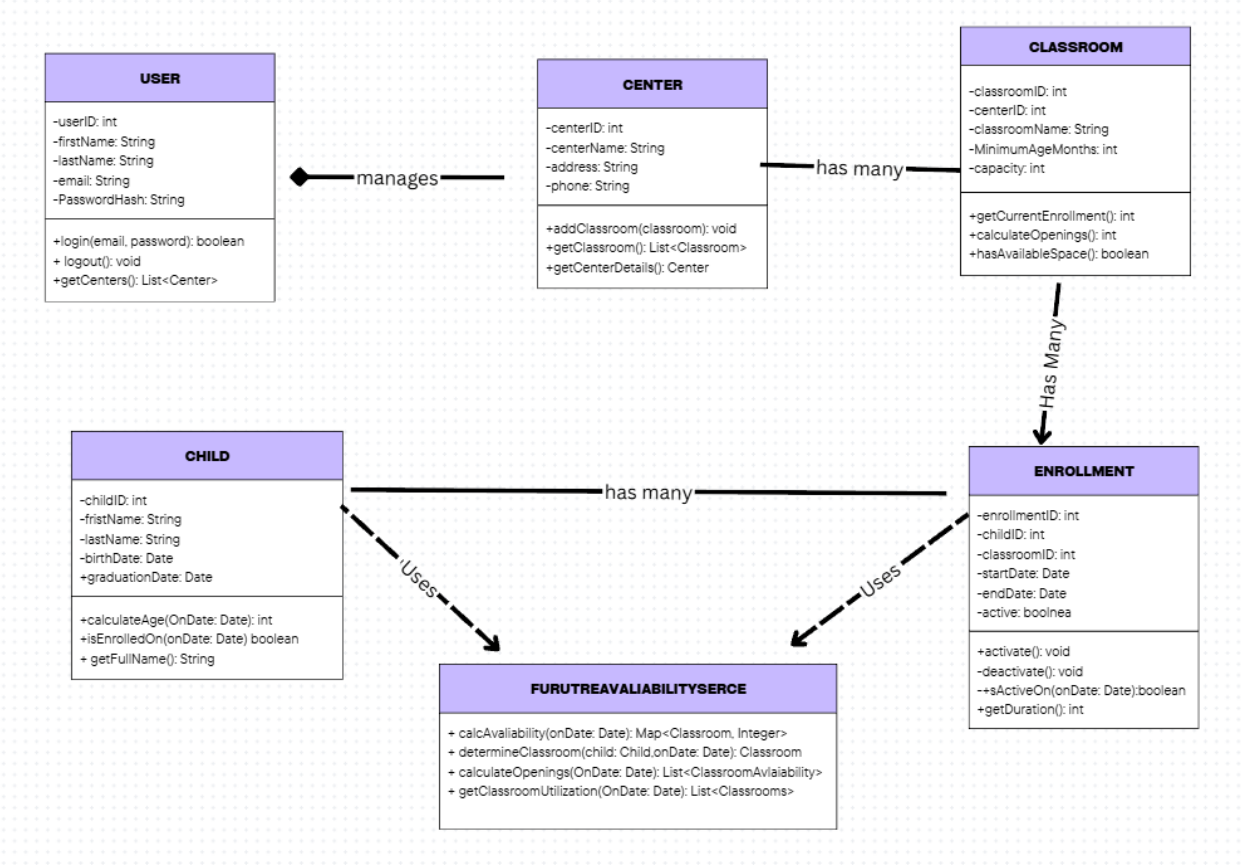
A Center contains multiple classrooms.

A Classroom contains multiple enrollment records.

A Child may have multiple enrollment records throughout their time in the system.

The EnrollmentService communicates with the FutureAvaliabilityService to calculate projected classroom openings.

Repository classes retrieve and update information stored within the MySQL database.



Process and Workflow Design

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The application follows a series of structured workflows to ensure enrollment information is processed consistently and accurately. Each workflow begins with user interaction through the web interface and ends with updated information being stored in the database or displayed back to the administrator.

Student enrollment Workflow

1. Administrator selects Manage Students
2. Administrator chooses Add Student
3. Child information is entered
4. Required fields are validates
5. Enrollment information is saved to the database.
6. The student list refreshed to display the new enrollment.

Future Enrollment Workflow

1. Administrator opens Future Enrollment
2. Administrator selects Future Enrollment or Month to Month
3. Administrator selects a projection date
4. The system retrieves all active enrollment records
5. The application calculates each child's age on the selected date.
6. Classroom assignments are determined based on age and classroom capacity.
7. Available openings are calculated for each classroom.
8. Students beginning within 5 weeks are added to the upcoming start preview
9. Students who have passed their projected exit dates are removed.
10. The system calculates classroom enrollment, capacity and openings.
11. Results are displayed using classroom lists and visual status indicators.

Algorithm Design

The Future Availability algorithm is the core business function of the system. Rather than only displaying current enrollment information, the application projects future classroom availability by evaluation each child's anticipated classroom placement.

The algorithm evaluates active enrollment records, calculates each child's projected age, determines the appropriate classroom based on business rules, verifies classroom capacity, and calculates availability openings for every classroom. This allows administrators to quickly determine future enrollment opportunities without manually reviewing student records.

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Future Availability Algorithm Pseudocode

```
BEGIN

User selects a projection date and forecasting view

Retrieve all active classrooms
Order classrooms from oldest to youngest

Retrieve all active students

FOR each student
   IF enrollment date is after the selected date
      Add student to Future Enrollments

      IF enrollment date is within five weeks
         Add student to the classroom upcoming-start preview
      END IF

      Continue to next student
   END IF

   IF selected date is after projected exit date
      Exclude student from active projection
      Continue to next student
   END IF

   Calculate student age on selected date
END FOR

Order eligible students from oldest to youngest

FOR each eligible student
   Determine age-appropriate classroom
```

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```
    IF age-appropriate classroom has capacity
        Assign student to that classroom
    ELSE IF a younger classroom has capacity
        Assign student to the younger classroom
        Mark student as held for capacity
    ELSE
        Add student to Students Needing Review
    END IF

    IF student becomes eligible for the next classroom
      within one month
        Mark student as approaching transition
    END IF
END FOR

Calculate enrollment, capacity, and openings
Display classroom rosters and future enrollment results

END
```

Business Rules

- Only active enrollments are included.
- Children are assigned based on age requirements
- Classroom capacities cannot be exceeded.
- If a classroom is full, the youngest eligible child remains in the previous classroom until space becomes available.
- Graduation dates remove children from future enrollment calculations.
- Students are processed from oldest to youngest when projected classroom placements are calculated
- A student that is old enough for the next classroom remains in the younger classroom when the next classroom is full.
- Students held back are identified in red.
- Students becoming eligible for the next classroom within a month are identified in yellow.
- Students starting within 5 weeks appear in the classroom at the bottom highlighted in blue

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Security Design

TinyTrack is a locally hosted admin application. The application does not currently include required user authentication or role-based authorization. These features will be added in the future.

Functional Requirements

Use Cases

TinyTrack's functional requirements describe the enrollment management and forecasting capabilities implanted in the final application. The following table summarizes those capabilities. The traceability matrix connects the documented requirements to application modules and test cases.

The Enrollment Management System provides the following major functional areas:

Functional Area	Description
Dashboard Management	Displays current enrollment, active classrooms, licensed capacity, and current openings
Student Management	Allows administrators to add, view, edit, search, assign, and inactive student records
Classroom Management	Allows administrators to add, view, edit, and inactivate classrooms and define age ranges and capacities
Center Management	Displays childcare center information and supports classroom relationships
Automatic Departure Calculation	Calculates August 25 of the year a student turns five as the projected departure date
Current Capacity Calculation	Calculates classroom enrollment and current openings using active enrollment data
Future Enrollment Projection	Projects future classroom enrollment for a selected date
Age-based classroom placement	Assigns students to projected classroom based on the selected date
Capacity-Constrained Placement	Prevents projected classroom placement from exceeding licensed capacity
Future-Start Enrollment	Includes students in projections beginning on their enrollment date

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Departure Processing	Removes students from projections after their projected departure date
Sample Data Loading	Loads fixed fictional student data without creating duplicated on repeated application startup
Month to Month Classroom View	Displays all projected classroom rosters together for a selected date
Students needing review	Identifies students who cannot be placed within the available licensed capacity

Reference: User Stories Attachment

Use Case or User Story	Approval Date	Justification
None	N/A	No functional requirements have been removed from scope at this time.

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Non-Functional Requirements

Use Cases

The following non-functional requirements define the operational characteristics of the Enrollment Management System.

ID	Requirement	Description
NFR-003	Future enrollment projections should return results within five seconds	The application should provide fast search results so administrators can efficiently answer enrollment inquiries
NFR-004	The application should function in current versions of Chrome and Edge	The system will function correctly in current versions of Google Chrome and Microsoft Edge without requiring additional software installation
NFR-006	The system validates required fields and preserves data integrity during creation and updates	The application will validate required fields and preserve accurate enrollment information whenever records are added, updated or removed.
NFR-007	The application provides user-friendly navigation and consistent styling	The interface will be intuitive and organized so administrators with varying levels of technical experience can easily complete common enrollment tasks.

No non-functional requirements have been removed from the scope at this time.

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Technical Requirements

Use Cases

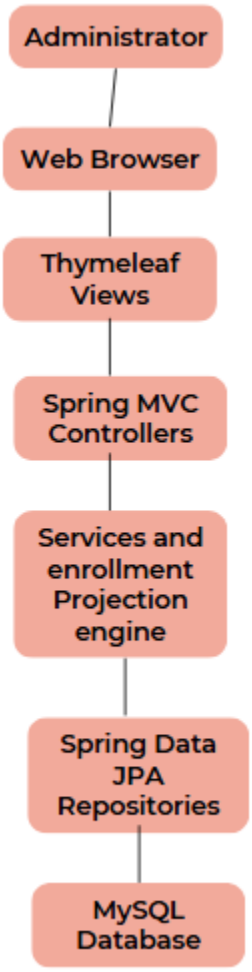
Below is a list of technologies and tools that will be used for this application.

Technology	Component
Java	Programming language
Spring Boot	Application framework
Spring MVC	Controller and web request handling
Spring Data JPA	Database access
Hibernate	Object-relational mapping
Thymeleaf	Server-side HTML templates
MySQL	Relational Database
HTML and CSS	User Interface
Maven	Dependency and build management
Git and GitHub	Version Control
VS Code	Development environment
MySQL Workbench	Database Management
Chrome and Edge	Browser testing
Windows	Development operating system

No technology has been removed from the scope at this time.

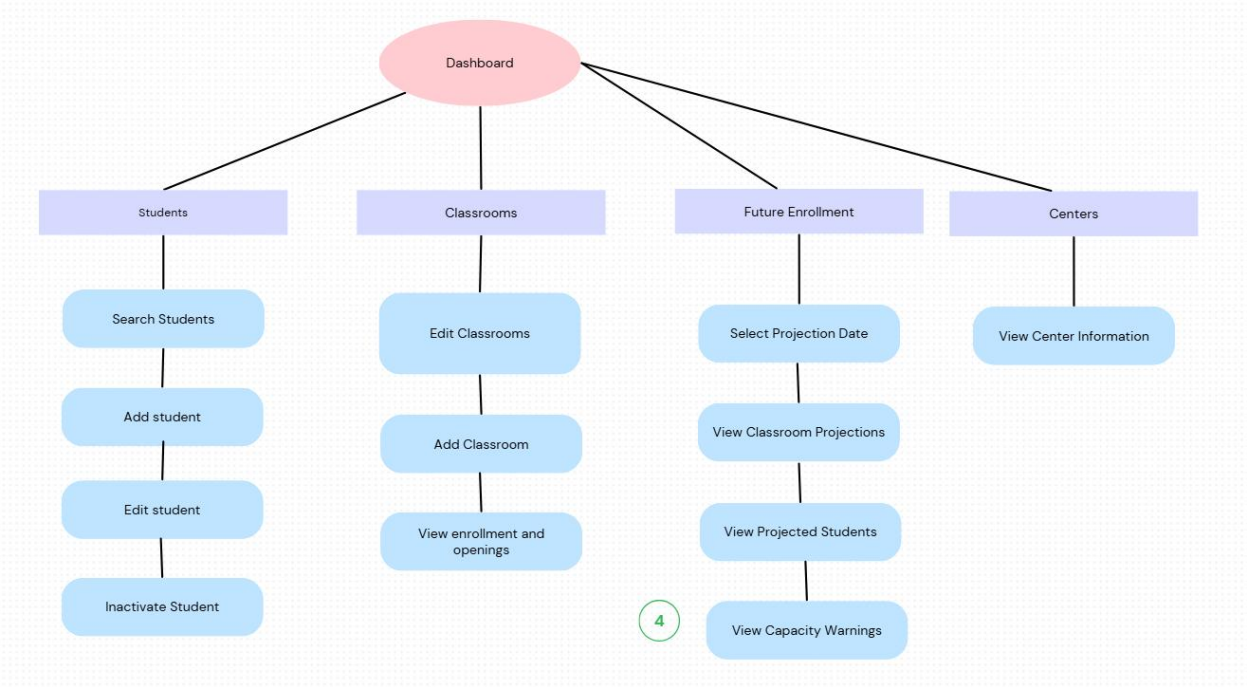
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Logical System Design



Site Map

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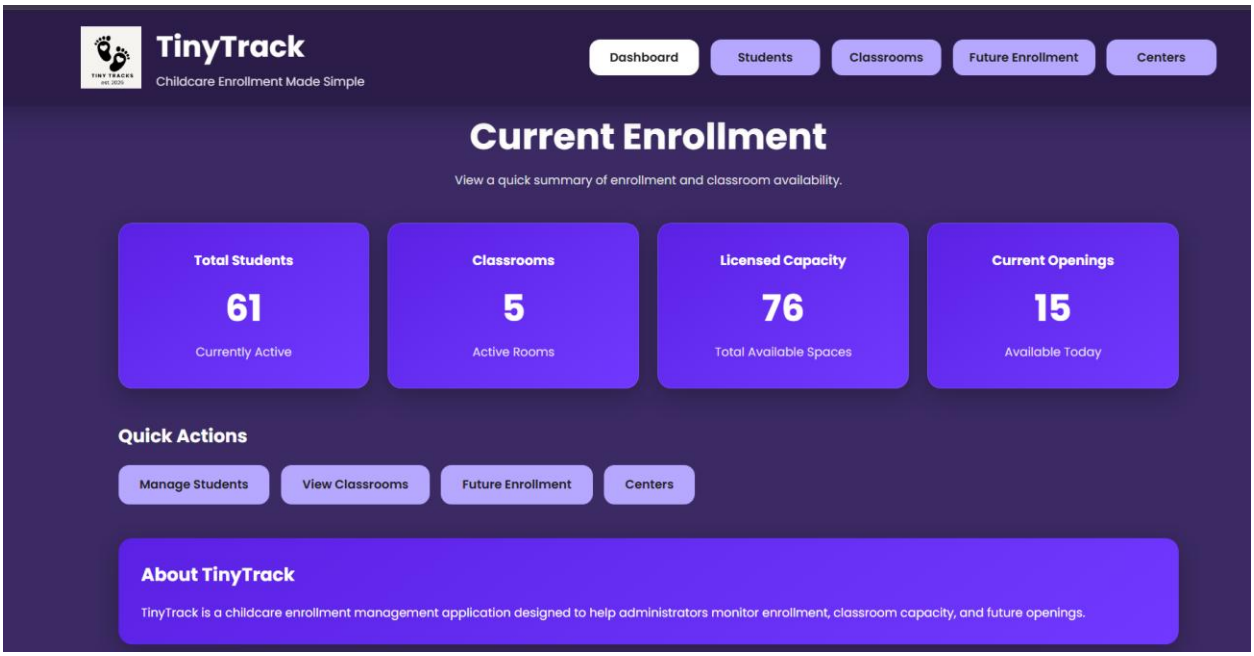
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User Interface Design

Below are images of the functioning TinyTrack application.

Home Page:

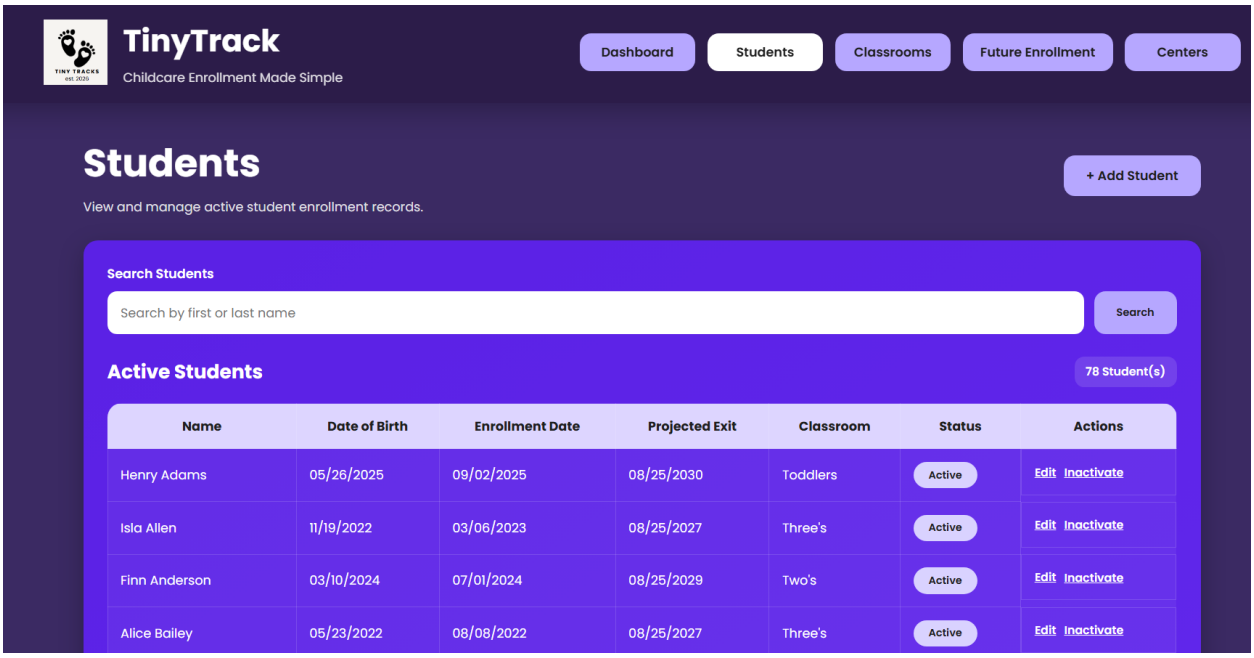
Below is the functioning home page. Displaying live data and functioning buttons.



Manage Students:

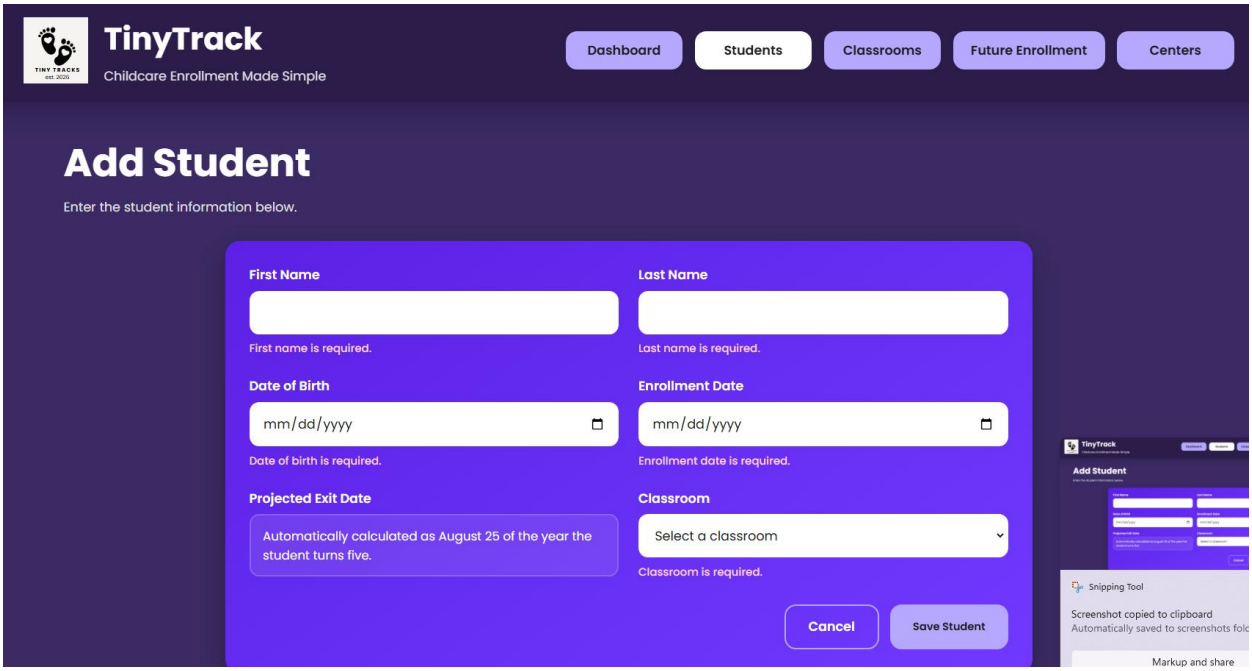
The student home page will display all active students with their DOB, enrollment date, Projected date, projected exit, classroom, status and actions to edit or inactivate. There is al the main navigation showing at the top and a button to add a student

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Add Student:

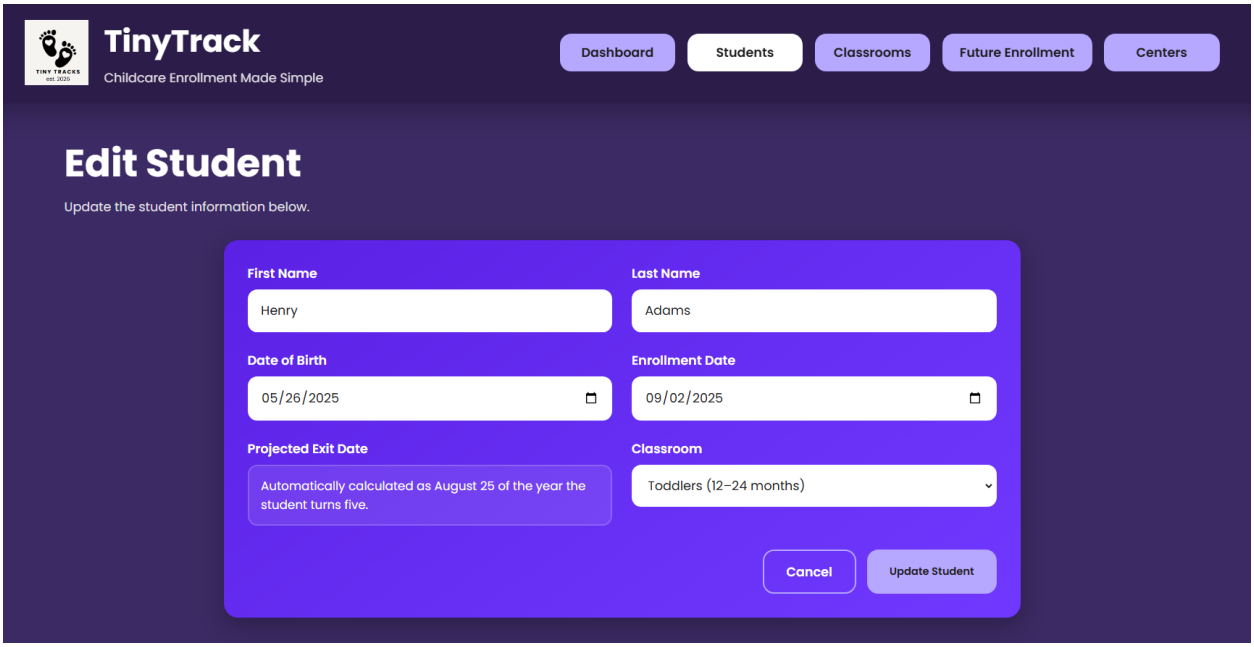
The add student page has a form to intake all data. There are also requirement checks for all data entered.



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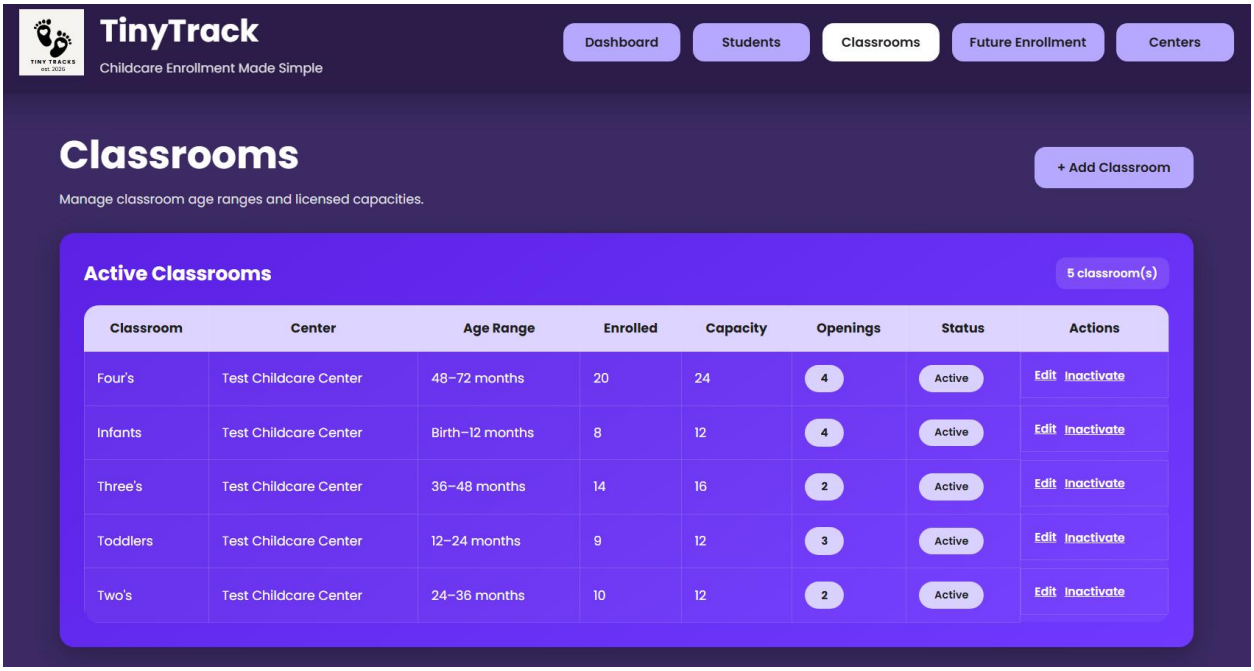
Edit Student:

Pulls the add student form with all information entered. At this time data can be changed and then the students information will be updated.



Manage Classrooms:

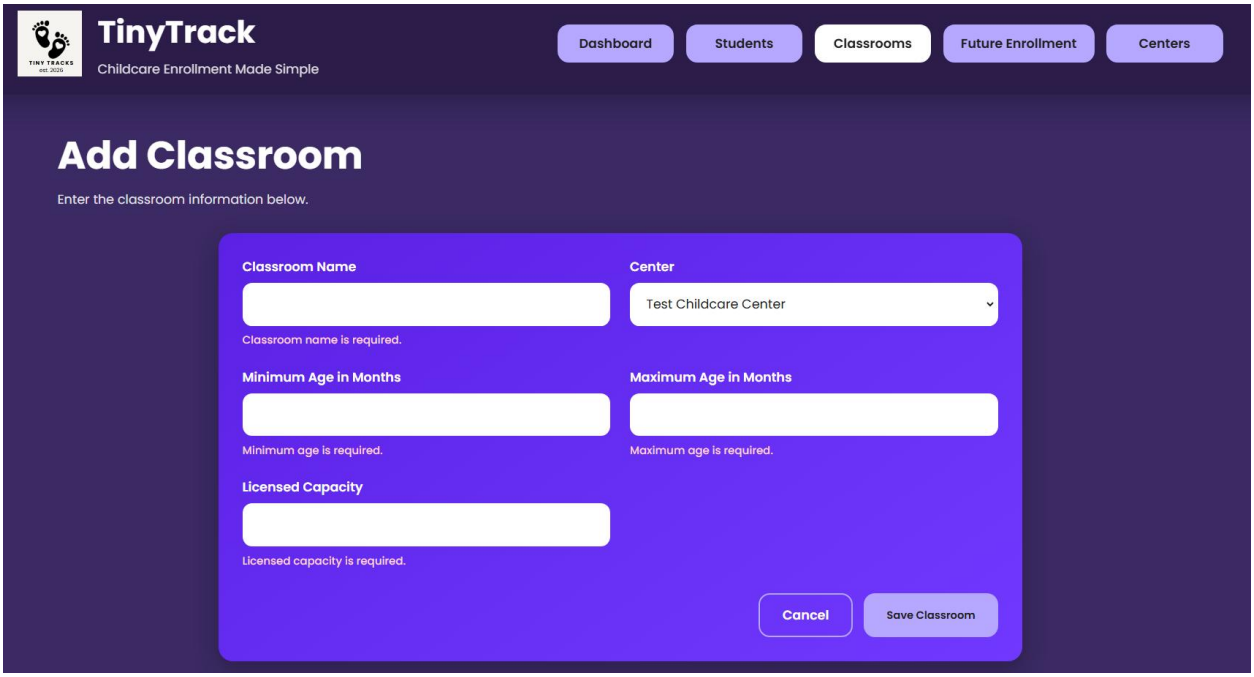
Classroom homepage shows all active classrooms along with their data. It also has a button to add classroom, edit and inactive classrooms.



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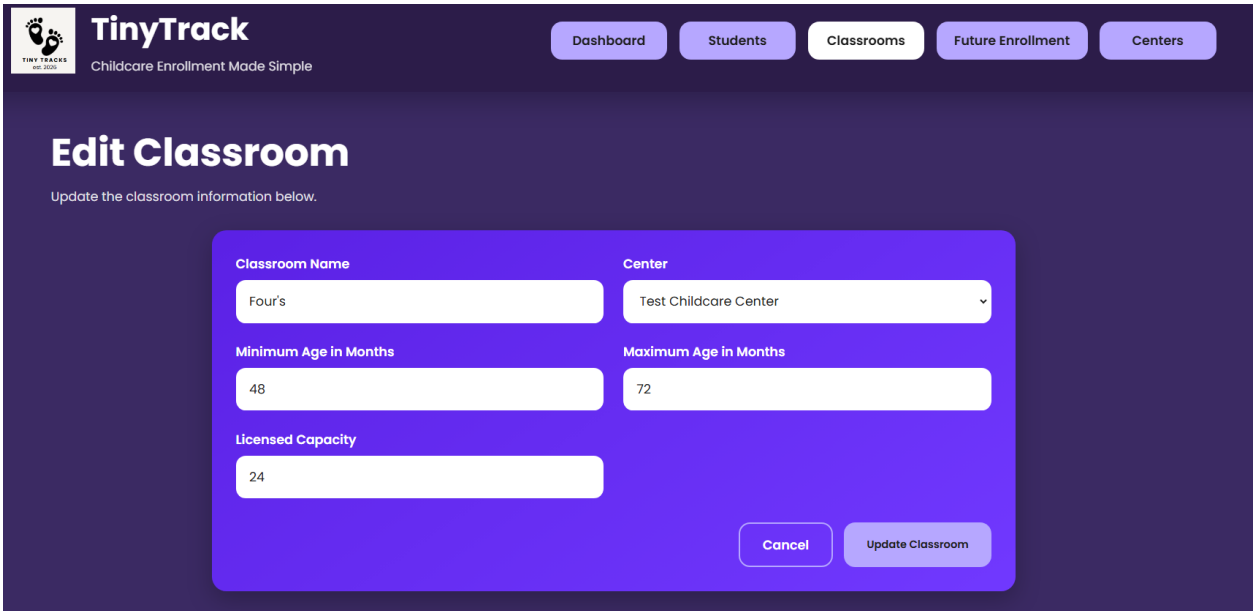
Add Classroom:

Add classroom displays a form with all data needed. It also checks that the data is correct and entered. The new classroom is then added to the Database.



Edit Classroom:

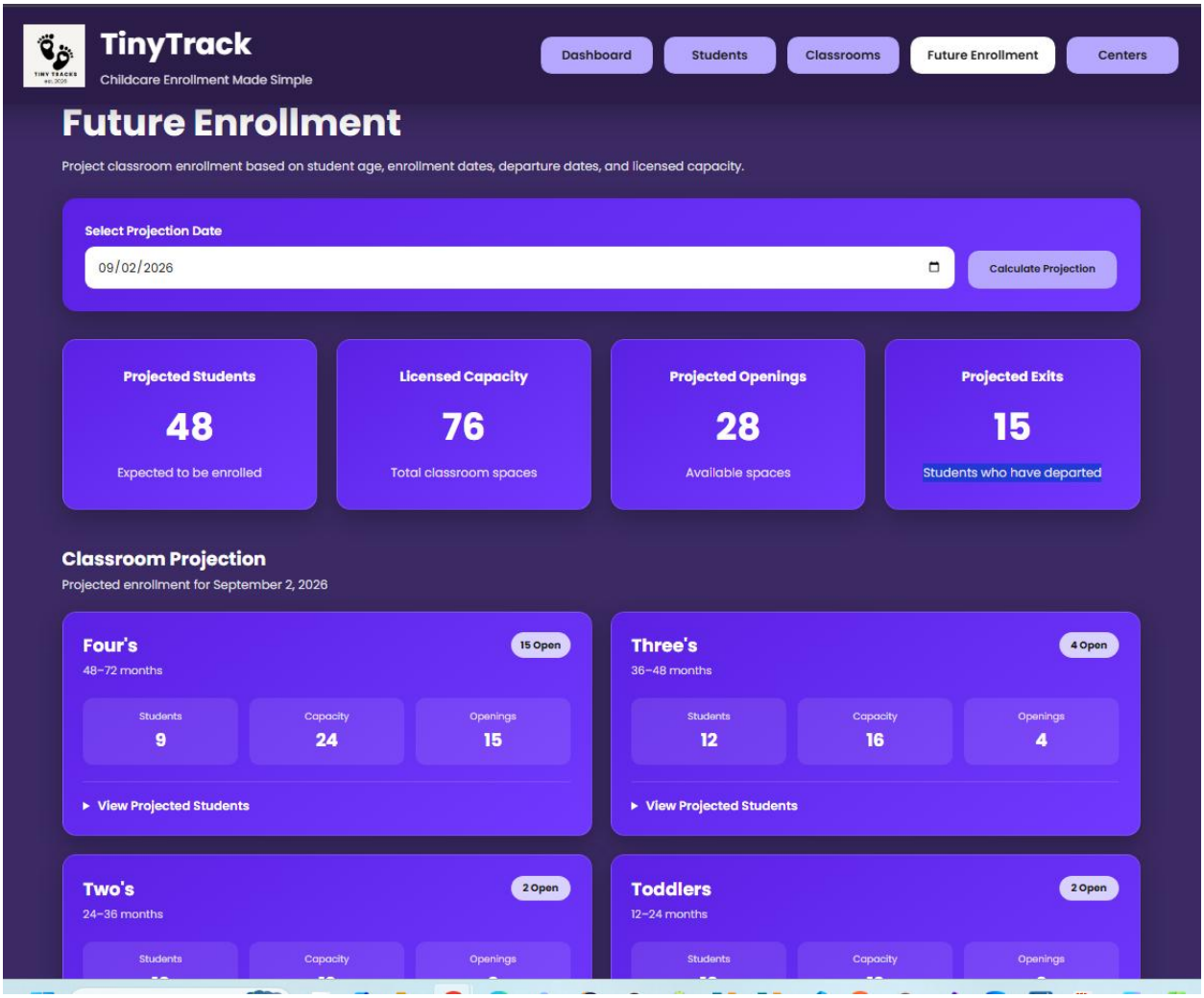
Edit Classroom displays the add classroom form with the data already entered. The user is then able to edit information and update it.



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Future Enrollment:

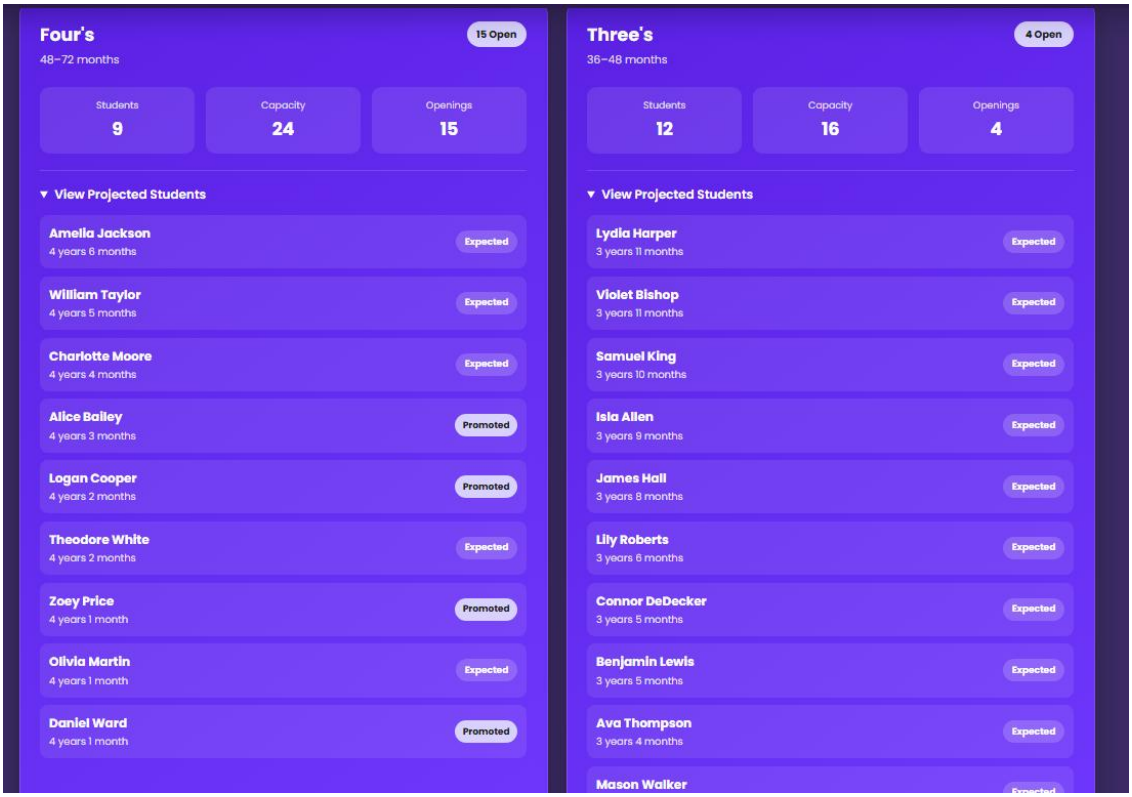
The future enrollment page automatically loads one month into the future and displays projected enrollment, licensed capacity, projected openings and projected exits.



Future Enrollment Classroom Breakdown:

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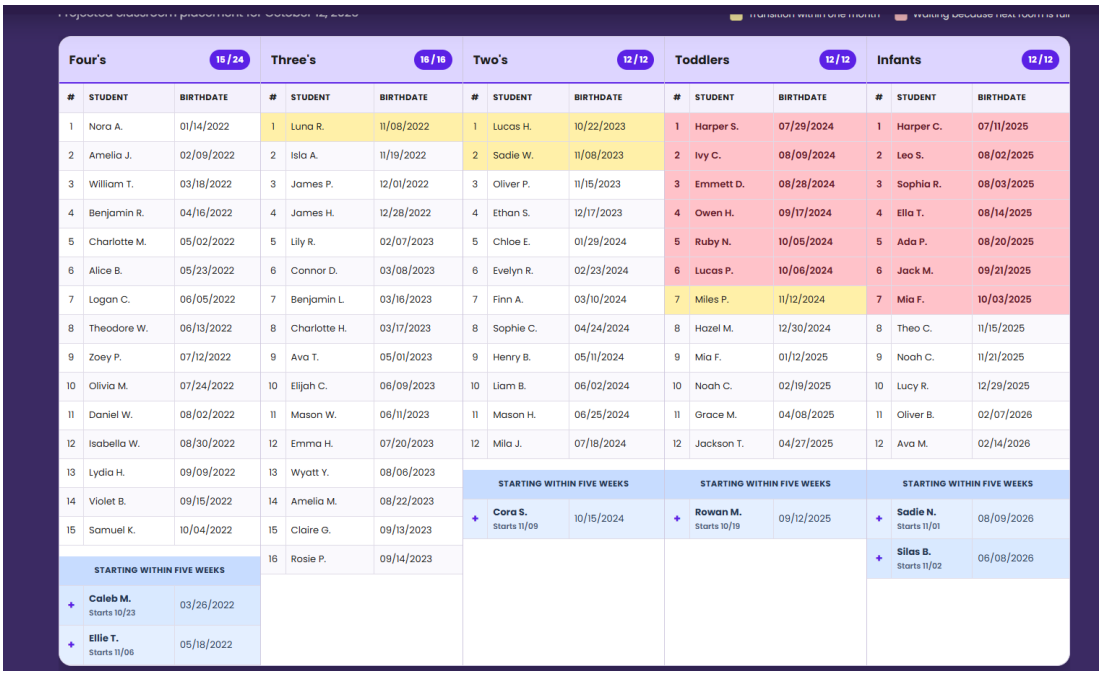
Here we see each classrooms capacity, openings and current students.



Month to Month classroom lists:

Here we see the month to month break down. In it we also see the upcoming students, students that were unable to move and students that will move soon.

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Reports and outputs

Report or output	Description
Dashboard Summary	Displays current students, active classrooms, licensing capacity, and current openings
Active Student List	Displays active students information, classroom assignment, enrollment date, and projected departure date
Student Search Results	Displays active students matching a first name or last name search
Classroom Capacity View	Displays classroom enrollment, capacity, and openings
Center Information	Displays childcare center contact and location information
Future Enrollment Projection	Displays projected classroom enrollment for a selected date
Projected Classroom Roster	Displays students assigned to each classroom for the selected future date
Projected Warnings	Identifies full classrooms, capacity holds, and students who cannot be placed

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Projected Exits	Displays the number of students no longer enrolled by the selected date
Month to Month Classroom Roster	Displays all projected classroom lists together for a selected date
Students needing review	Displays students who cannot be placed within licensed classroom capacity

TinyTrack's final implementation provides student and classroom record management, center information, current enrollment calculations, automatic projected departure dates, and future classroom forecasting. The projection engine considers age, enrollment dates, departure dates, and licensed capacity. The forecasting interface displays classroom rosters, approaching transitions, capacity holds, upcoming starts, future enrollment schedules, and students needing review.

The application uses Java, Spring Boot, Thymeleaf, Spring Data JPA, and MySQL. The final submission documents the implemented functionality and identifies remaining limitations and planned improvements.

Planned Improvements

Future development may include administrator authentication, role-based permissions, waitlist management, CCA and private-pay tracking, expanded multi-center administration, and production hosting. These capabilities are not included in the final implementation.

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